

Hero inverted-U — three model layers (v1)

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Audience: Charles + Alex read-through
Canonical hero folder: `sports/datasets/mbb/exports_inverted_u_v0/alex_side_by_side_v0/`
Related: `05_Model_Nesting_Note_v1.md`, `30_Alex_Gates_inform_status_outline.md`, `BINDING_Selection_is_its_own_step.md`, `re_entry/02_Three_Kinds_of_Model.md`

0. BINDING — environment ≠ advancement; score ≠ select

Do not merge “describe the peer environment” with “who wins the scarce slot.” Inside advancement, **do not call the score “selection.”**

Mechanism	Object	Question
Environment	<code>L_net = B(.) - D(.)</code>	How do peers help vs hurt (development, visibility, crowding around you)?
Scoring	Alex <code>S_i = A_i - λ·L_C</code>	How do we rank candidates? (λ lives here.)
Selection	Winner rule (v1: top K)	Given ranks, who wins the scarce slot?

Unified nest: `S_i = A_i + λ(B - D) = A_i - λ·L_C` when `(B - D) = -L_C` for the ranking channel. Knockout: `λ = 0` ⇒ `S_i = A_i`.

- **Hero (Layer A):** outcome only — draft rate vs pool quality. Does **not** identify environment vs advancement channel.
- **540 / Pass A (Layer C):** makes **score** → **select** explicit — rank by score → top K; knockout = talent-only score vs congestion-in-score.

The hero describes outcomes; `L_net` is the peer environment; advancement = score then select; the sim tests congestion in the score under a fixed winner rule.

1. Locked hero estimand (fixed target for all tests)

All Layer A/C comparisons use this empirical spec unless explicitly noted as robustness.

Field	Value
Outcome	<code>Y_draft</code> (ever drafted), mean rate per bin
X-axis	<code>poolq_loo</code> (LOO mean teammate <code>perf</code>)
perf	ESPN box ppm, z-scored within collegiate season
Bins	16 quantile (<code>poolq_binning='quantile'</code>)
Winsor	<code>poolq_loo</code> at (0.01, 0.99) before binning
min_minutes	20 (applied before LOO when rebuilding from box)
Panel source	<code>use_prebuilt_panel_csv=False</code> for canonical rebuild path
Draftee filter	Off (<code>restrict_teams_by_draftees=False</code>)
Seasons	2011–2021
Sample (Jul 2026 lock)	62,180 player-seasons; 1,134 drafted

File naming slug: `empirical_ppm_poolq_loo_16quantile_winsor0199_min20_2011`

Shape (qualitative): rise through middle ventiles → plateau bins 12–15 (~2.3–2.6%) → **dip at bin 16 only** (~1.2%).

2. Three layers — do not merge into one “model”

Layer	Object	Repo anchor	What a “pass” proves	What it does not prove
A. Phenomenological	$Y \sim \beta_0 + \beta_1 \cdot \text{poolq_loo} + \beta_2 \cdot \text{poolq_loo}^2$	<code>530</code> / <code>panel_build.draft_poolq_quadratic_coeffs</code>	Inverted curvature in the real panel on the hero estimand	Mechanism; that NBA uses this score
B. Structural (theory)	Three parts: (1) <code>L_net = B - D</code> environment; (2) Alex <code>S_i</code> scoring; (3) winner rule select	Nesting note §2–3; BINDING	Environment help/hurt and D may enter the score ; select is separate	Separate estimation of B(Q) and D(Q) in v1; hero ≠ scoring equation
C. Generative (sim)	Assign (<code>p</code>) → score (<code>λ</code>) → select (top K)	<code>540_READ_ME_SIM.md</code> , <code>540_three_step_sim.ipynb</code> , <code>tier1*.py</code>	Congestion in the score changes who gets selected; talent-only score fails	Bin-for-bin reproduction of empirical <code>poolq_loo</code> ventiles

Alex side-by-side (v1): Layer A hero PNG + Layer C sim PNG on same Y (advancement rate), honest X (empirical LOO quality vs sim LOO bins), plus limitation sentence below.

3. How each layer is tested

Layer A — regression on real rows

- Filter panel to hero estimand → OLS $Y_{\text{draft}} \sim 1 + \text{poolq_loo} + \text{poolq_sq}$ (LPM = linear probability model, not linear programming).
- Pass: $\beta_2 < 0$ (fitted curve **concave down** / inverted-U tendency); optional overlay tracks ventile bin means.
- What the quadratic buys you: a signed curvature claim (β_2) plus one turning point $Q^* = -\beta_1 / (2\beta_2)$ (local max if $\beta_2 < 0$). **Not** inflection points (need richer than quadratic). Claims are about the **fitted** curve; hero **bins** stay the stylized fact. Not mechanism.
- Not required:** R^2 , causal interpretation, or match to sim.

Layer B — prose + ingredient knockouts

- No single estimated formula in v1.
- Part 1:** B – D frames the **environment** (help vs hurt among peers).
- Part 2:** Alex score frames **ranking** — congestion enters the **score**, not “the whole model.”
- Part 3:** Top K (or later stochastic draw) is the **winner rule**.
- Pass:** Narrative consistency + sim shows $\lambda=0$ (ability-only **score**, same top-K) **fails** the generative congestion story.

Layer C — simulation

- Draw synthetic league → **assign** rosters (ρ) → **score** → **select** top K → bin by **poolq_loo** (16 quantile).
- Pass A (talent-only score):** `score_mode='ability'` / $\lambda = 0$ → **monotone** selection rate on **poolq_loo** (bin 1 ~0.7% → bin 16 ~28%); no elite dip.
- Pass A (congestion in score):** `score_mode='loo_gap_plus_ability'`, `loo_pool_l_mode='crowding_smooth'`, `w=0.5` → elite-tail **compression** (bin 16 ~16% vs ~28% talent-only); not pointwise hero match.
- Pass B (ρ ablation):** low / moderate / high assortativity + sort-and-chop diagnostic → `alex_rho_ablation_v0/` (assignment plumbing, not the headline theorem).
- Scripts:** `sports/scripts/hero_model_reset_bundle.py` (Pass A); `sports/scripts/540_rho_ablation_bundle.py` (Pass B).
- Archived labs:** 538D etc. under `sports/archive/` — historical source; daily workspace is **540**.

4. Simplification strategies (both are valid)

Strategy	Use for	Procedure
Bottom-up	Layer A	Start quadratic → decompose Q only at Rung 2.5 (<code>poolq_loo</code> vs <code>crowding_smooth</code> in CELL 5d)
Top-down	Layer C	Full generative stack → zero λ / ability-only score / change ρ → record what breaks

Use **bottom-up** first for the hero; **top-down** for 540 mechanism strip-down.

5. Simulation plumbing vs core claim

Knob	Role	v1 gate?
λ / w on L_C in score	Core mechanism	Yes (Pass A done)
Top-K selection	Scarcity (winner rule)	Yes (conceptual)
Soft assignment (ρ , T_j)	Realistic pools	Helpful, not theorem (Pass B diagnostic)
Sort-and-chop overlay	Hard assortative diagnostic	No — not $\rho \rightarrow \infty$
Preferential attachment	Boolean, default off	No — stretch
Faithful historical team sweeps	Realism stretch	No — see §6
Bin-for-bin LOO match to hero	North-star R&D	No — see §6

Notation: User-facing assignment knob is ρ . Legacy τ (temperature) appears only in archived docs. Soft crowding inside `L_C` ≠ assignment softness ≠ selection noise.

Path II (locked): Generative figure may bend on LOO `poolq_loo` bins while empirical hero is the **replicated fact** on the same conditioning object — still **not** claiming pointwise match without a dedicated calibration pass.

6. Explicitly deferred (stretch, not v1 success criteria)

Do **not** block Alex draft or side-by-side on:

- Bin-for-bin replication** of hero ventile draft rates from sim on `poolq_loo`.
- Rivanna / faithful** full sweeps until minimal score→select story stays locked.
- Separate B(Q) and D(Q) estimation** on one axis.
- Dense assortativity grids**, preferential attachment on, empirical team cloning, HS binning, multiplicative rewrite.

These remain **parallel R&D** after the minimal POC bundle is frozen.

7. Limitation sentence (Alex slide / manuscript §5)

*The empirical stylized fact is defined on **leave-one-out pool quality** among teammates; the minimal generative proof-of-concept **scores** with a term that penalizes leave-one-out viable-peer congestion, **selects** top K, and plots binned **selection rates on the same LOO quality axis**. We do **not** claim bin-for-bin reproduction of the empirical ventile draft rates in v1; we claim that **talent-only scoring is insufficient** and that **congestion in the score** can bend advancement curves in a disciplined artificial league.*

(Source: nesting note §4; shortened for slides.)

8. Keepers vs reorder (Jul 2026 reset · sync Jul 28)

Keep: `530` pipeline, hero triplet, nesting note, BINDING, re_entry 00–03, **540** three-step sim, Pass A/B exports, Pertinent Thoughts sensitivities, `alex_side_by_side_v0/` / `alex_rho_ablation_v0/`.

Reorder claims: Phenomenon first (hero) → minimal score ingredient (λ) → theory prose — not “one formula does everything.” Keep **score ≠ select** in every write-up.

Deliverables from this reset: this memo, `lpm_hero_coefficients.txt`, generative knockout CSVs, `inverted_u_side_by_side_empirical_vs_generative.png` in `alex_side_by_side_v0/`, p ablation bundle in `alex_rho_ablation_v0/`.